Ahmed Aly

Senior Mobile Developer

eng.ahmed.ally2@gmail.com | 01225108720 | 01021081942 | Cairo / Egypt | github
linkedin medium

Profile

Senior Mobile Developer with 4 years of professional experience designing, building, scaling, and maintaining production mobile applications across **Native Android** (Kotlin, Java) and **Flutter** (Dart). Strong experience with Jetpack Compose, MVI, MVVM, BLoC, Clean Architecture, multi-module codebases, Coroutines, Flow, Dart Streams, Ktor, Retrofit, Dio, and dependency injection. Experienced in real-time communication with Socket.IO, offline-first persistence, resilient networking, background processing, maps and live location services, mobile security, automated testing, code quality, and CI/CD automation using Fastlane. Focused on maintainable architecture, reliable user experiences, and scalable feature delivery across the mobile development lifecycle.

Professional Experience

Senior Mobile Developer - SHIFT inc 01/2024 - present | Cairo, Egypt

- Developed and maintained SHIFT Inc. (Native Android) and contributed to SHIFT Driver using Flutter.
- Built scalable Android features using Kotlin, Jetpack Compose, MVI, Clean Architecture, and multi-module architecture.
- Developed Flutter driver workflows using Dart, BLoC, Cubit, Provider, GetIt, Dio, and local caching.
- Integrated Google Maps, routing, geofencing, live GPS tracking, and real-time communication using Socket.IO.
- Implemented resilient networking with Ktor Client and Circuit Breaker using fail-fast and fallback strategies.
- Improved background processing using Foreground Services and WorkManager, and automated releases with Fastlane and FCM.

Mobile Developer - SURE International Technology 01/2023 - 12/2023 | Cairo, Egypt

- Developed and maintained Android and Flutter applications for multiple client projects in a software-house environment.
- Built scalable features using Kotlin, Java, Dart, Jetpack libraries, Flutter, MVVM, BLoC, Provider, and Clean Architecture.
- Integrated REST APIs and real-time communication using Retrofit, Dio, Firebase, and Socket.IO.
- Collaborated with cross-functional teams, fixed complex bugs, and improved application performance, stability, and maintainability.

Projects

SHIFT - Carsharing & Chauffeur

Customer mobility application supporting car-sharing bookings and chauffeur ride workflows, with maps, live trip updates, offline persistence, background processing, and push notifications. Built with Kotlin, Jetpack Compose, MVI, Coroutines, Ktor Client, Socket.IO, Room, DataStore, WorkManager, Maps SDK, FCM, TLS, and Fastlane.

SHIFT Driver (Flutter Cross-Platform)

Driver-facing application for chauffeur operations with real-time dispatching, live GPS tracking, navigation, offline persistence, camera and QR workflows, and push notifications. Built with Flutter, Dart, BLoC, Cubit, Provider, Freezed, GetIt, Dio, Drift, Hive, Secure Storage, Socket.IO, Geolocator, Google Maps, GoRouter, FCM, BLoC Test, Mocktail, and Fastlane.

SHIFT Logistics

Last-mile logistics application supporting driver attendance, route optimization, shipment barcode scanning, delivery monitoring, electronic proof of delivery (E-POD), and vehicle-condition reporting to support daily logistics operations.

Fleet Management

Internal fleet-operations application for vehicle inspections, readiness checks, vehicle-status workflows, and remote car lock and unlock operations, supporting day-to-day fleet readiness and control.

Skills

Native Android Development

Languages & UI (Kotlin, Java, Jetpack Compose, XML)

Architecture & Design (MVI, MVVM, MVP, Clean Architecture, Onion Architecture, Multi-Module Architecture, Repository Pattern, SOLID)

Concurrency & Reactive (Kotlin Coroutines, Flow, StateFlow, SharedFlow, RxJava)

Networking & Real-Time (Ktor Client, Retrofit, OkHttp, Socket.IO, REST APIs, Circuit Breaker, Chucker)

Persistence & Offline (Room Database, DataStore, SharedPreferences, Offline-First Architecture, Caching Strategies)

Dependency Injection (Hilt, Dagger 2, Koin)

Background Processing (Foreground Services, WorkManager)

Location & Device APIs (Google Maps, Location Services, Geofencing, Geocoder, Camera, Barcode/QR Scanning)

Testing & Code Quality (JUnit, MockK, Mockito, Turbine, Coroutine Test, Espresso, ktlint, Android Lint, Detekt, LeakCanary, StrictMode)

Security (Android Keystore, HTTPS/SSL/TLS, Certificate Pinning, Secure Storage, AES/RSA Encryption, Secure Key Generation)

Cross-Platform Development (Flutter & Dart)

Core (Flutter, Dart, Futures, Streams, Async/Await, Null Safety)

State Management & Architecture (BLoC, Cubit, Provider, Riverpod, Clean Architecture, Feature-First Modularization, Repository Pattern)

Data Modeling & Code Generation (Freezed, JSON Serialization, Build Runner, Immutable State Models)

Networking & Real-Time (Dio, REST APIs, Socket.IO)

Dependency Injection & Navigation (GetIt, Injectable, GoRouter, Deep Linking)

Persistence & Offline (Drift/SQLite, Hive, Flutter Secure Storage, Shared Preferences, Offline Caching)

Native Integration (MethodChannel, EventChannel, Native Android/iOS Bridging, Platform Views)

UI & Device Features (Custom Painters, Slivers, Animations, Geolocator, Camera, Barcode/QR Scanner, WebViews)

Testing (Flutter Test, BLoC Test, Mocktail, Widget Testing, Integration Testing, Golden Tests, Dart Analyzer)

Mobile Engineering & DevOps

CI/CD & Automation (Fastlane, CI/CD Pipelines, Firebase App Distribution)

Version Control (Git, GitHub, Bitbucket)

Firebase (Firebase Cloud Messaging, Firebase Authentication)

Engineering Practices (Code Review, Clean Code, Performance Profiling, Modularization, Maintainable & Scalable Codebases)

Education

Bachelor's Degree in Computer Science 09/2016 - 11/2020 | Zagazig, Egypt
Faculty of Science, Zagazig University

Languages

Arabic
Native

English
Upper Intermediate (B2)